Negative Literature Answer to the Bounded Multiplier Question in arXiv:1206.3358

FA/Banach run packet

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Status

This is a literature-already-answered packet. It records that the bounded Fourier multiplier equality question from Chen–Xu–Yin [1] was answered negatively in the later paper of Ricard [2]. It is not an original proof from this run.

Original Problem

In Section 7, after proving the theta-independence of completely bounded L_p Fourier multipliers, Chen–Xu–Yin ask:

> Let $2 < p \leq \infty$. Does one have
> $$\mathrm{M}(L_p(\mathbb{T}^d_\theta)) = \mathrm{M}(L_p(\mathbb{T}^d))?$$

They conjecture a negative answer and give only an infinite-generator $p = \infty$ illustration.

Supporting Answer

Ricard [2] explicitly starts from the Chen–Xu–Yin completely bounded result and states that bounded multipliers behave differently. In the arXiv source, Theorem `ex1` proves that for any irrational θ and any $1 < p \neq 2 < \infty$ there is $\phi : \mathbb{Z}^2 \to \mathbb{C}$ such that

$$\|M_\phi : L_p(\mathbb{T}^2) \to L_p(\mathbb{T}^2)\| < \infty \quad \text{but} \quad \|M_\phi : L_p(\mathbb{T}^2_\theta) \to L_p(\mathbb{T}^2_\theta)\| = \infty.$$

This disproves the equality for every $2 < p < \infty$.

For the endpoint $p = \infty$, Proposition `ex3` in [2] constructs, for every irrational θ, a symbol $\phi : \mathbb{Z}^2 \to \mathbb{C}$ such that

$$\|M_\phi : L_\infty(\mathbb{T}^2_\theta) \to L_\infty(\mathbb{T}^2_\theta)\| < \infty \quad \text{but} \quad \|M_\phi : L_\infty(\mathbb{T}^2_\theta) \to L_\infty(\mathbb{T}^2_\theta)\|_{cb} = \infty.$$

Ricard recalls immediately before this proposition that the latter cb norm agrees with the classical $L_\infty(\mathbb{T}^2)$ multiplier norm by the Chen–Xu–Yin theorem. Hence the endpoint equality is also false.

Literature Answer. *The equality*

$$\mathrm{M}(L_p(\mathbb{T}^d_\theta)) = \mathrm{M}(L_p(\mathbb{T}^d))$$

asked in [1] has a negative answer in the finite-dimensional quantum torus setting for every $2 < p \leq \infty$. *It already fails for irrational two-tori.*

Literature identification. For $2 < p < \infty$, apply Ricard's Theorem `ex1` with $d = 2$ and irrational θ. The displayed symbol is a bounded classical multiplier but is not bounded on the corresponding quantum L_p-space, so the two multiplier spaces cannot be equal.

For $p = \infty$, apply Ricard's Proposition `ex3`. The constructed symbol is bounded on $L_\infty(\mathbb{T}^2_\theta)$ but has infinite completely bounded norm there. Since the completely bounded norm is equal to the classical $L_\infty(\mathbb{T}^2)$ multiplier norm, the same symbol is not a bounded classical multiplier. Thus the endpoint equality also fails. □

Scope Limitations

This packet records a known negative answer for irrational two-dimensional quantum tori. It does not classify rational theta or the exact structure of the bounded multiplier spaces.

Verification Notes

The source and supporting papers are distinct. The supporting paper is a later paper whose abstract and main results explicitly distinguish bounded multipliers from the completely bounded theta-independent result of Chen–Xu–Yin. The local duplicate search covered arXiv ids `1206.3358` and `1512.01142`, plus phrases `bounded Fourier multipliers`, `quantum tori`, `M(L_p(T_theta))`, and `Ricard`.

References

[1] Z. Chen, Q. Xu, and Z. Yin, *Harmonic analysis on quantum tori*, arXiv:1206.3358; Commun. Math. Phys. 322 (2013), 755–805.

[2] E. Ricard, *L_p-Multipliers on Quantum Tori*, arXiv:1512.01142; J. Funct. Anal. 270 (2016), 4604–4613.